

Group Assignment

Group 11

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1. Problem Identification and Initial Dataset

Problem Definition: This project aims to build a supervised machine learning model that predicts whether a startup will be successful or unsuccessful based on key attributes such as funding, revenue, valuation, customer base, age and industry. The goal is to provide investors and founders with a bias-aware, data-driven decision-support tool for evaluating startup potential across different sectors.

Startups play a crucial role in innovation-driven economies, yet many fail to achieve profitability despite significant funding. Predicting the likelihood of profitability and overall success can help investors, venture capitalists, and founders make better-informed decisions regarding funding allocation, risk assessment, and strategic planning.

The goal of this project is to develop a machine learning model that can analyse startup attributes to predict whether a startup is likely to be successful. The is expected to provide actionable insights into factors that most influence success.

Business Objective

The business objective is to assist decision-makers in evaluating startup potential using historical data. By predicting success based on characteristic like business sector, age, funding amount, revenue, valuation, number of customer, the model can:

- Support venture capital investment decisions
- Help startups benchmark themselves against successful peers.
- Enable data-driven risk assessment in early-stage investment portfolios

Dataset Source

The dataset used for this analysis is the **Global Startup Success Dataset**, available publicly on **Kaggle** <https://www.kaggle.com/datasets/hamnakaleemds/global-startup-success-dataset>

The dataset includes startup-level information such as company name, year of establishment, country, industry sector, funding amount, valuation, stage, and profitability status. The target variable i.e. *Success_Flag* indicates whether a startup has achieved success or not.

While searching for data related to start-ups, we found two datasets:

1. *startup_data.csv*
2. *global_startup_success_dataset.csv*

We opted for *global_startup_success_dataset.csv* because *startup_data.csv* contains only 923 rows, which is too small for effective training and testing.

Dataset Overview

The dataset contains information on 5,000 startups across multiple industries, countries, and business stages. It includes both categorical and numerical variables that describe each startup's geographical orientation, age, financials, operational characteristics, and outcomes.

Justification for Dataset Selection

This dataset was chosen because:

- It provides realistic, multi-dimensional data relevant to modern business and investment decision-making.
- It enables exploration of classification techniques in machine learning.
- It contains numerical, categorical features, making it ideal for demonstrating data preprocessing, feature engineering, and model tuning processes.
- The data is well-structured and clean, which supports effective EDA and modelling within a business analytics context.

2. Exploratory Data Analysis

The purpose of the Exploratory Data Analysis (EDA) phase was to examine the structure, quality, and underlying patterns within the startup dataset before feature engineering and model development. This stage involved inspecting dataset structure, validating completeness, analysing distributions of key variables, exploring correlations between financial and performance metrics, and conducting deeper industry-level correlation comparisons.

2.1 Dataset Overview

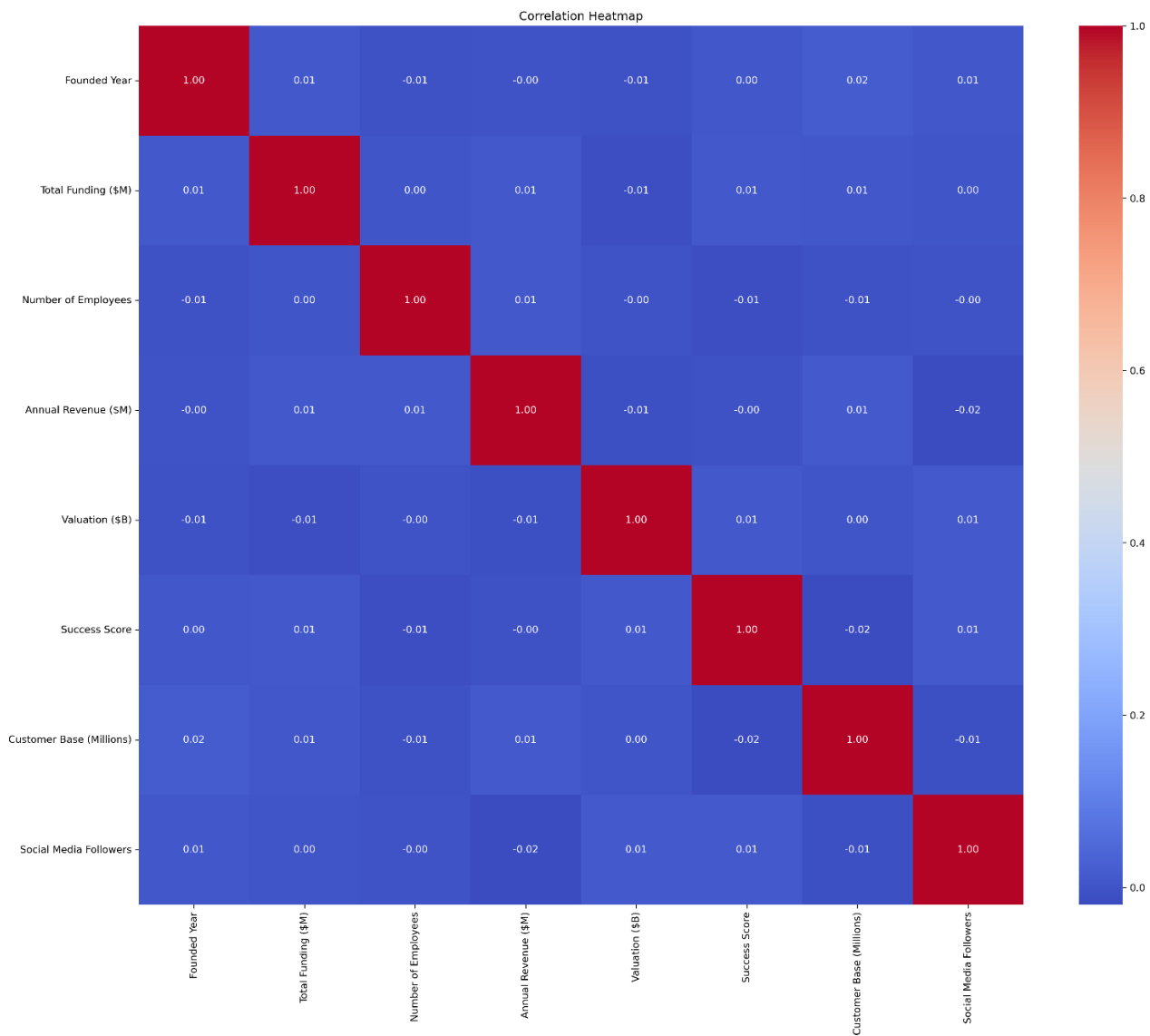
The dataset was first inspected using `df.head(10)` and `df.info()` [as shown in #2.3] . This initial review confirmed that the dataset contains:

SL	Column Name	Description	Data Type	Unique Values
1	Startup Name	Name of the startup. Serves mainly as an identifier and has not been used for prediction	<i>object</i>	5,000
2	Founded Year	The year in which the startup was established	<i>int64</i>	23
3	Country	Country where the startup is headquartered	<i>object</i>	10
4	Industry	Industry sector of the startup (e.g., FinTech, HealthTech, EdTech)	<i>object</i>	10
5	Funding Stage	Stage of funding (e.g., Seed, Series A, Series B, Series C, IPO)	<i>object</i>	5
6	Total Funding (\$M)	Total amount of funding received in millions of USD	<i>int64</i>	499
7	Number of Employees	Number of employees working in the startup	<i>int64</i>	3,167
8	Annual Revenue (\$M)	Reported annual revenue in millions of USD	<i>int64</i>	990
9	Valuation (\$B)	Market valuation of the startup in billions of USD	<i>float64</i>	3,131
10	Success Score	Composite score (0–10) indicating overall startup performance	<i>int64</i>	9
11	Acquired?	Whether the startup has been acquired (Yes/No)	<i>object</i>	2
12	IPO?	Indicates if the company has gone public (Yes/No)	<i>object</i>	2
13	Customer Base (Millions)	Approximate number of customers in millions	<i>int64</i>	99
14	Tech Stack	Category describing the startup's primary technology platform (e.g., AI, Cloud, SaaS)	<i>object</i>	5
15	Social Media Followers	Total number of social media followers across platforms	<i>int64</i>	4,998

The `df.info()` output confirmed that all 15 features had complete data with no missing values or duplicates. Therefore, no data cleaning or imputation was necessary before proceeding with the analysis.

2.2 Correlation Heatmap

The correlation heatmap [as shown in #2.4] revealed that all numerical features exhibit very weak correlations, with values ranging only from -0.02 to 0.02 . This means that none of the financial variables such as Total Funding, Annual Revenue, Valuation, or Customer Base have any strong linear relationship with each other. Importantly, the Success Score also showed no correlation with any other metric, confirming that it is not a meaningful performance indicator. Overall, the dataset demonstrates a flat correlation structure, which is consistent with a synthetic dataset constructed to avoid multicollinearity.



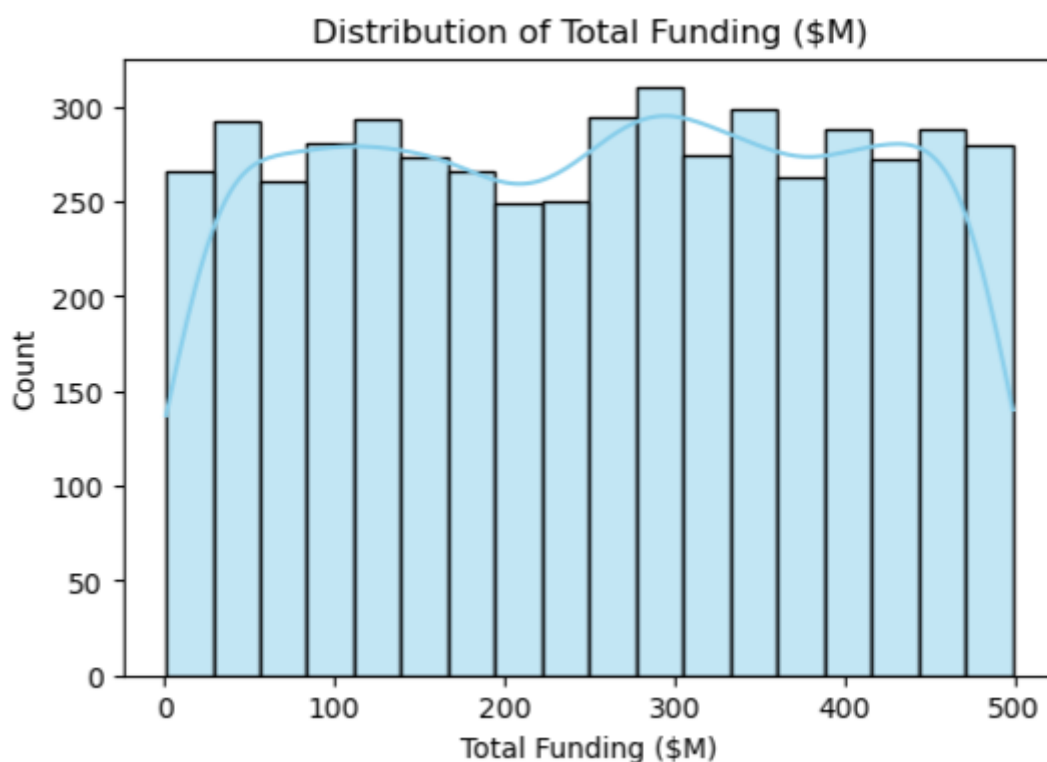
2.3 Analysis of Categorical Features

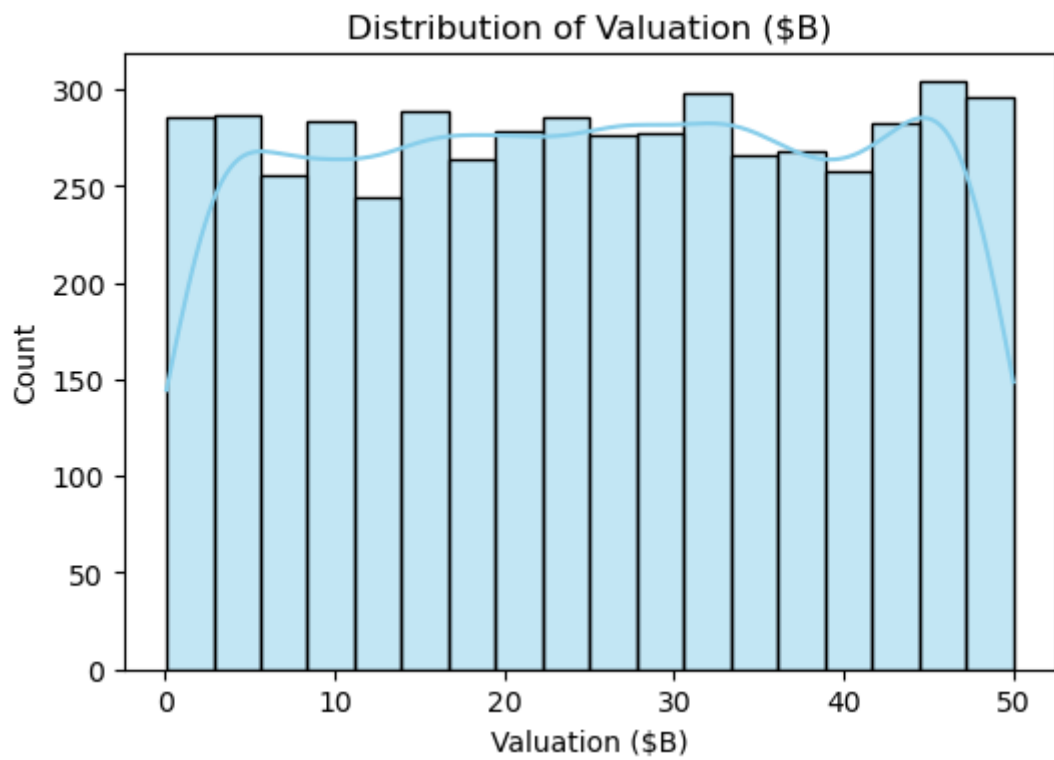
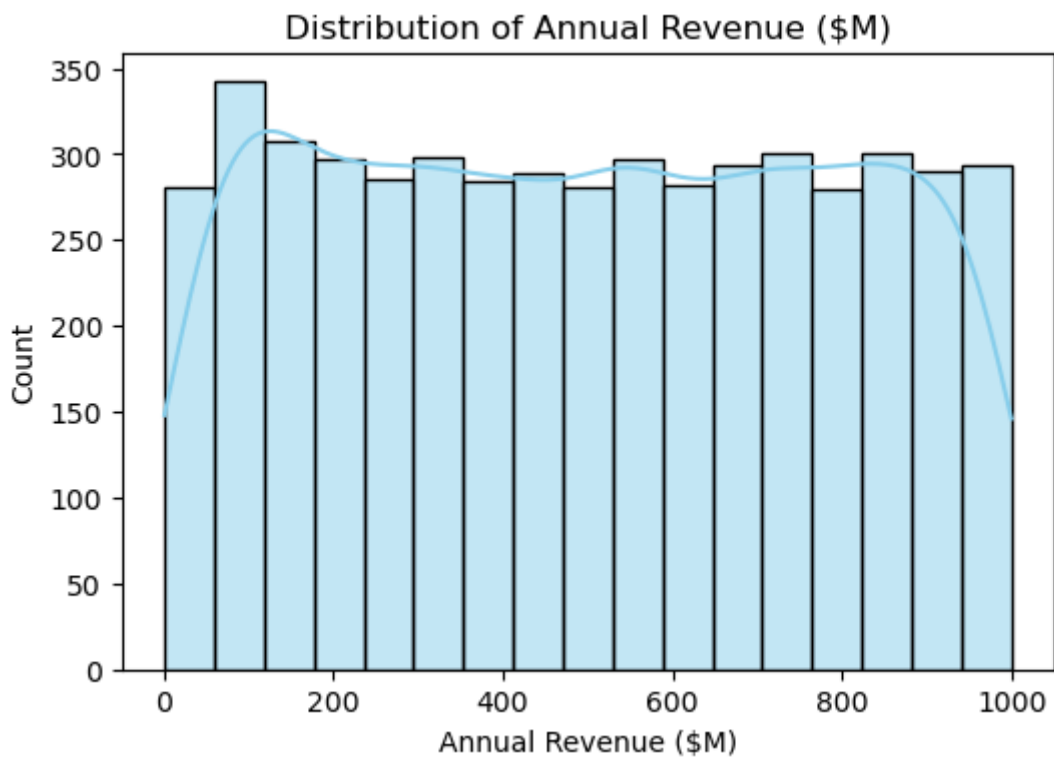
The categorical feature analysis demonstrated that the dataset is well balanced across countries, industries, and funding stages. The country distribution [#2.5 and #2.7] showed relatively even representation, with Germany (539), Brazil (519), Japan (502), the UK (501), and the USA (501) appearing most frequently, while the remaining countries each ranged between 478 and 498 entries. A similar pattern was observed in the industry distribution [#2.6 and #2.9], where categories such as Tech (524), Gaming (520), EdTech (518), Energy (510), and Logistics (509) exhibited comparable counts, and all other industries also fell within the 470–500 range. This balanced structure ensures that no single country or sector disproportionately influences the analysis. Additionally, the funding stage distribution [#2.8] showed a healthy spread across Seed, Series A/B/C, and IPO stages, indicating a diverse set of startup maturity levels within the dataset. Overall, the categorical variables reflect a well-constructed and evenly distributed dataset suitable for unbiased modelling.

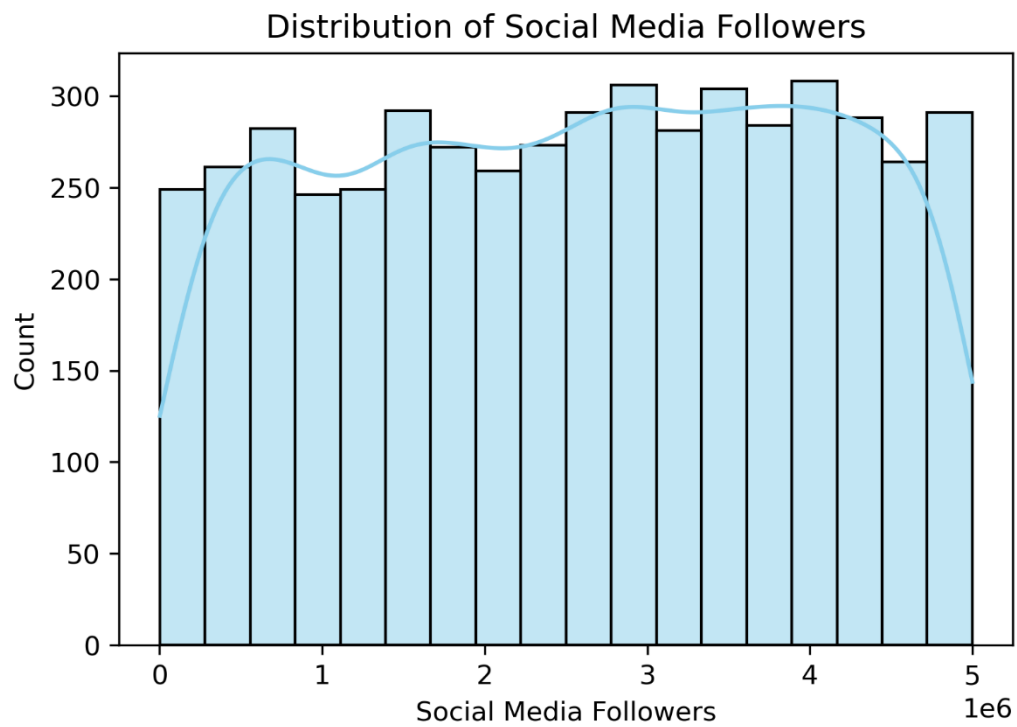
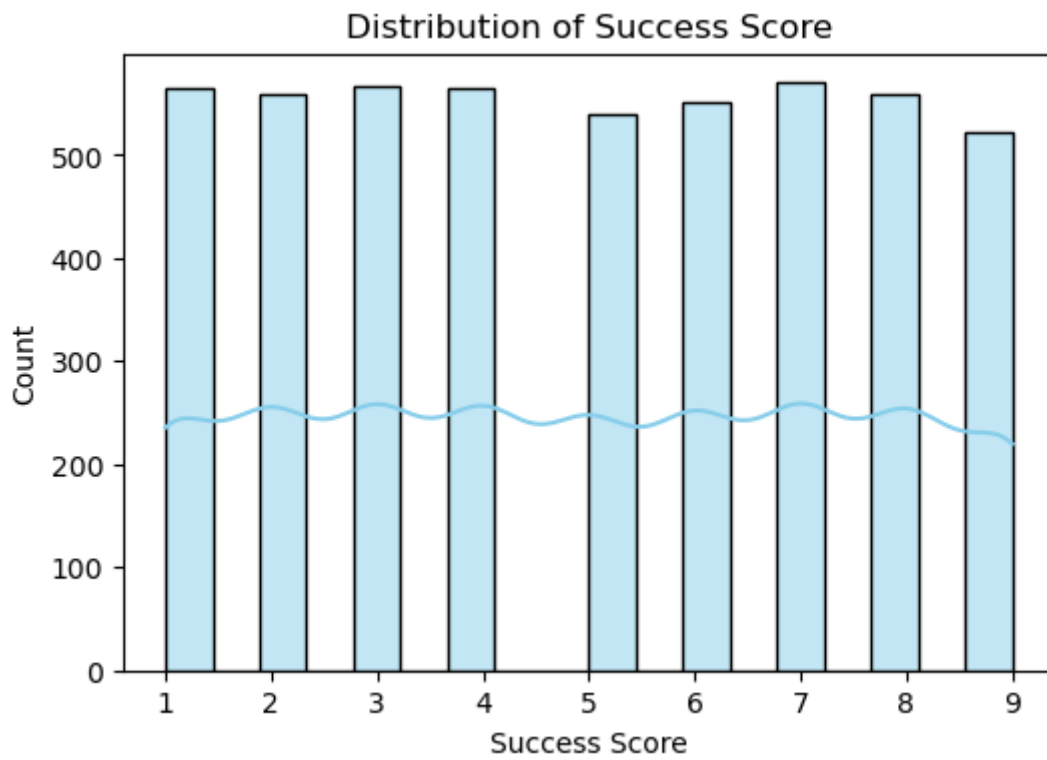
2.4 Numerical Feature Distribution

Histograms and boxplots [#2.10] explored the distributions of:

1. Total Funding (\$M)
2. Annual Revenue (\$M)
3. Valuation (\$B)
4. Success Score
5. Social Media Followers







Key observations:

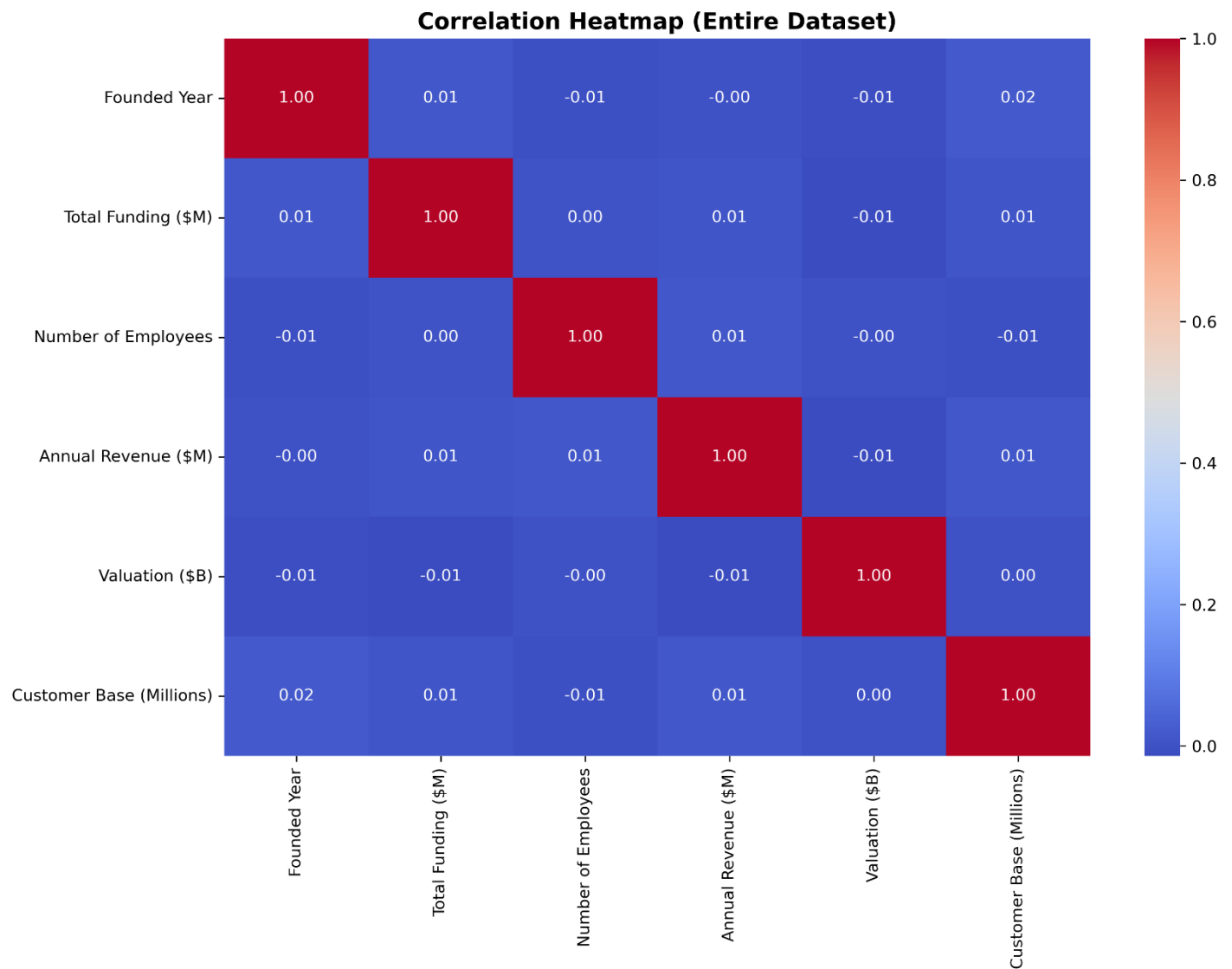
- All numerical distributions appear fairly uniform.
- There is no significant skewness or heavy-tailed behaviour.
- No extreme outliers were detected.
- The dataset behaves consistently with a synthetic, uniformly generated structure.

2.5 Correlation Analysis (Full Dataset)

A correlation heatmap [as shown in #2.14] demonstrated that overall correlations between core numerical variables are extremely low, for example:

- Total Funding ↔ Valuation: 0.01
- Annual Revenue ↔ Customer Base: 0.00
- Success Score ↔ Total Funding: 0.01

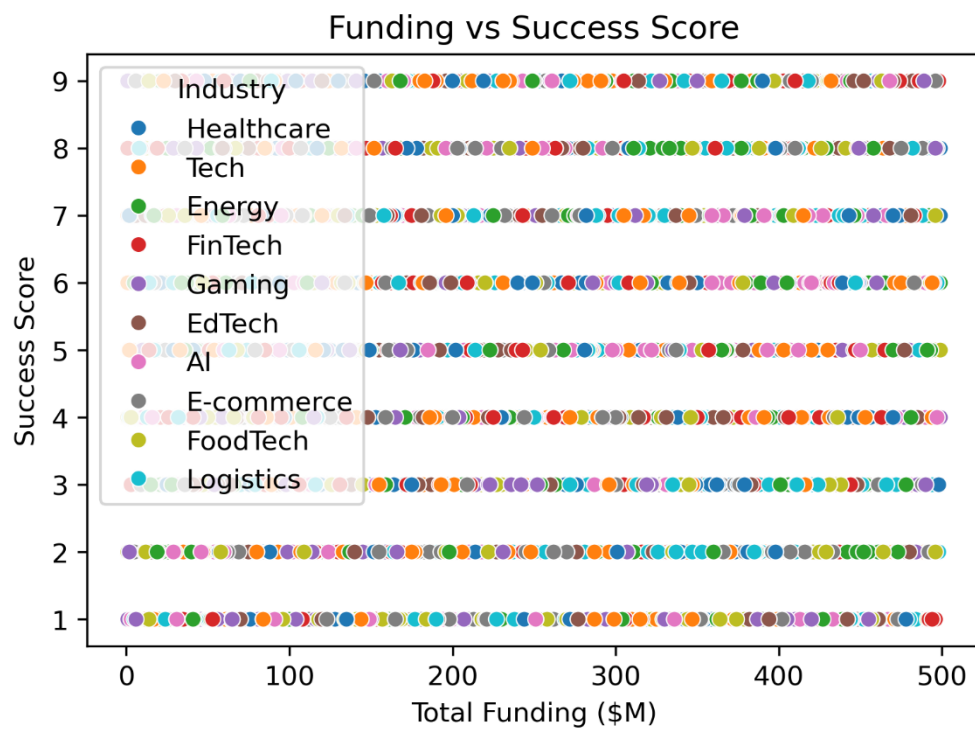
The Success Score variable showed no meaningful relationship with any other metric. This confirmed that the Success Score is arbitrary and not suitable as a performance target.



2.6 Visual Exploration of Key Relationships

2.6.1 Total Funding vs Success Score

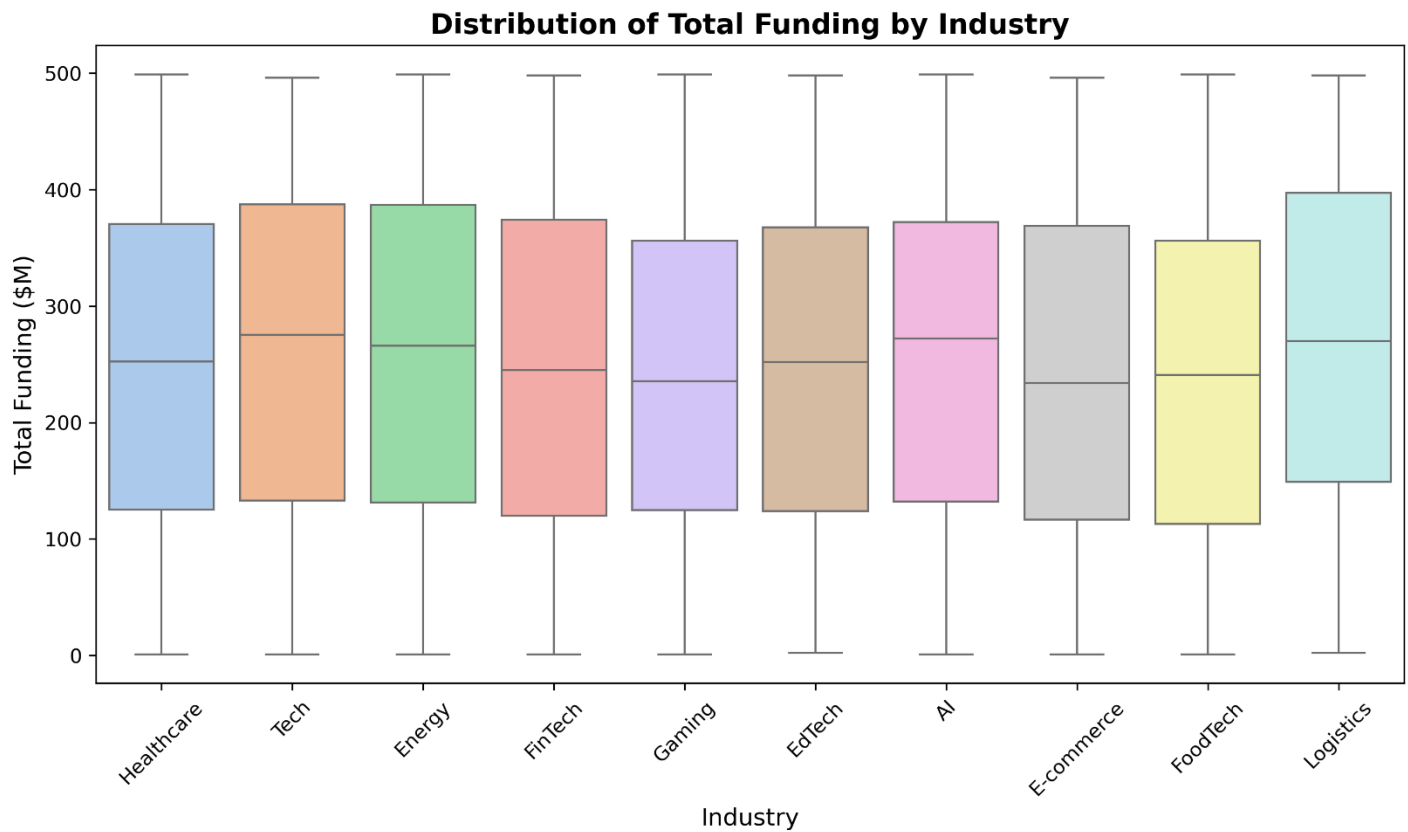
The scatterplot [#2.12] showed no visible pattern or trend between Total Funding and Success Score. This reinforced that Success Score should not be used for predictive modelling.



2.6.2 Funding Differences Across Industries

A boxplot [#2.13] highlighted differences in funding levels:

- Tech, Logistics, and Energy showed slightly higher median funding.
- FoodTech and Healthcare exhibited greater variability in funding.
- AI and FinTech were more stable.



These patterns supported the later use of industry-normalised metrics, as industries differ structurally.

2.7 Profitability Ratio Analysis

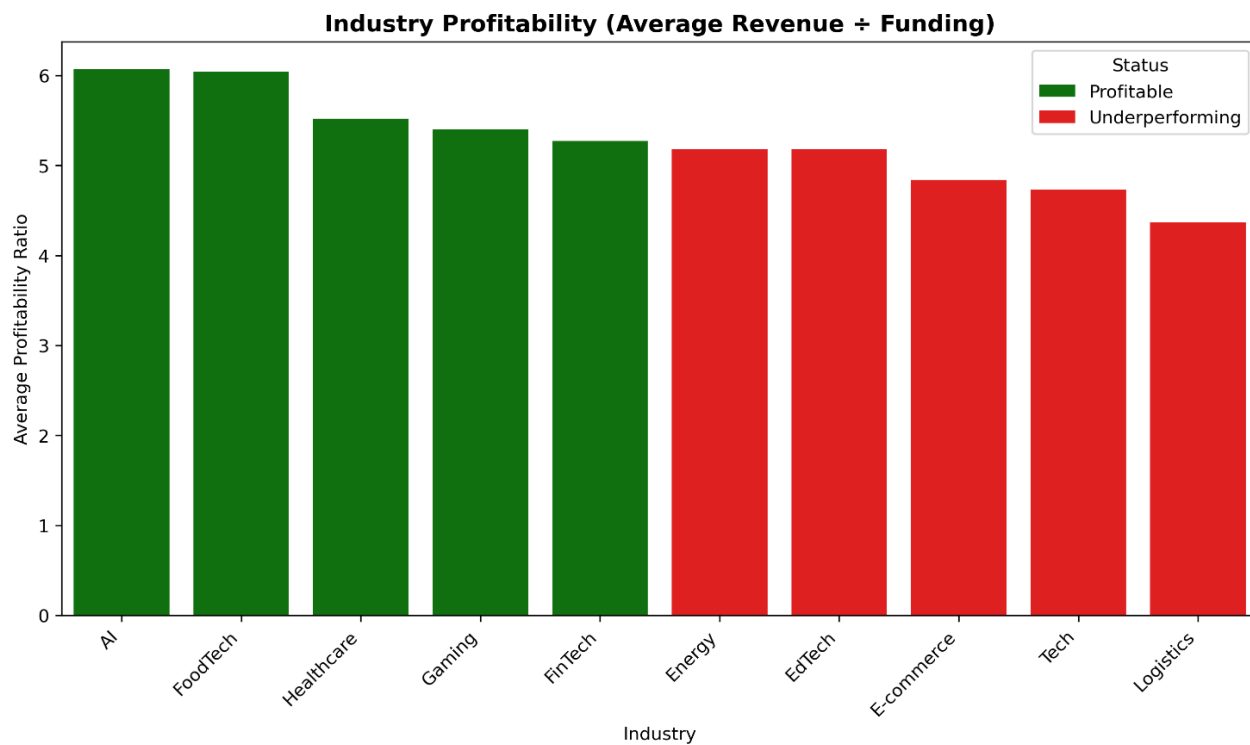
A Profitability Ratio was computed [#2.15]:

$$\text{Profitability Ratio} = \frac{\text{Annual Revenue}}{\text{Total Funding}}$$

After clipping extreme values, industry averages were calculated (pages 16–17):

- Highest profitability: AI (6.07), FoodTech (6.04)
- Lowest profitability: Logistics (4.37)

This indicates that industries differ in how efficiently they convert funding into revenue, highlighting the need for normalisation when defining performance metrics.



s 2.8 Per- Industry Dataset Splits

As shown in #2.16, the dataset was split into separate CSV files per industry.

This enabled detailed correlation comparisons within each industry.

Industry-Specific Correlation Differences

Industry-level correlation matrices [#2.16] were compared with the overall dataset to identify whether industries behaved differently. Although small variations appeared for example:

- Slightly stronger valuation–revenue correlation in AI
- Slightly weaker funding–valuation correlation in Gaming

- Minor increases in profitability correlations in FinTech

The differences are extremely small (typically between 0.01 and 0.08).

This means:

- No industry shows strong or meaningful deviation from the global pattern.
- Differences are statistically negligible.
- Results reflect minor fluctuations, consistent with a synthetic dataset intentionally designed with low correlations.

Therefore, while each industry shows slightly different patterns, the effect sizes are too small to carry real interpretive significance.

The practical takeaway is simply that industry-normalisation is appropriate not that industries behave fundamentally differently.

2.8 Summary of Key EDA Implications

The findings from the EDA provided several important insights that shaped the subsequent stages of the project. First, the Success Score variable was removed, as it showed no meaningful correlation with any performance-related metrics and did not effectively distinguish successful from unsuccessful startups. Additionally, several features such as Startup Name, Funding Stage, Acquired?, IPO?, and Tech Stack were identified as low-contribution variables with limited predictive value and were therefore excluded from further analysis. The EDA also highlighted notable differences across industries in terms of funding, revenue, valuation, customer base, and profitability ratios, reinforcing the need for industry-level normalisation to ensure fair comparison between startups. The profitability ratio proved valuable in revealing how efficiently different industries convert funding into revenue, informing the development of more meaningful performance indicators. Finally, although industry-specific correlation differences were small, they still supported the decision to engineer a composite, industry-aware success metric (Success_Flag) that would more accurately capture relative performance within each sector.

3. Initial Data Preparation

Initial data preparation focused on removing low-value columns, encoding categorical variables and constructing a more meaningful binary success indicator prior to loading the data to any machine learning algorithm.

3.1 Feature Selection and Removal

Following columns were removed because they contributed little predictive value or risked introducing noise:

- Startup Name – A pure identifier with no predictive meaning
- Country – weak correlation with success and potentially confounded by other features.
- Funding stage: A descriptive maturity indicator that do not carry significant meaningful predictive value.
- Acquired? and IPO? – Sparse binary indicators that overlapped conceptually with success itself.
- Tech Stack – Highly categorical and fragmented with no direct influence
- Success Score – A composite metric that did not correlate meaningfully with other features.

3.2 Encoding Categorical Variables

Industry, a categorical variable, was converted to numeric representation using one-hot encoding.

3.3 Creating Age and The Target Variable

- A new column, Age, was created to normalize startup maturity across industries and provide a consistent comparative feature for the predictive model
- The Success Score variable was eliminated, and in its place, a new binary target variable called Success_Flag was created that classified startups as either successful (1) or unsuccessful (0). In order to be equitable to all the different sectors, all the important numerical features Valuation (\$B), Annual Revenue (\$M), Customer Base (Millions), Total Funding (\$M), and Age were normalized in their corresponding industries. Each feature was divided by the mean of the particular industry in order to minimize bias that comes from differences in the scale of the industry. After normalization, a weighted composite performance score was determined using the predefined weights which showed up each feature's importance Valuation (0.25), Total Funding (0.25), Customer Base (0.20), Annual Revenue (0.20) and Age (0.10). This composite score displayed the total performance of each startup. Surpasses were made at last, implementing percentile-based threshold, whereby above the 75th percentile of their industry the values got designated as successful (1) and those below this threshold were labeled as unsuccessful (0). The creation of Success_Flag was preceded by the following code which helped generating these normalized features and computing the composite score.

3.4 Scaling Numerical Features

StandardScaler was applied to all numeric inputs to address differing feature scales (e.g., *Valuation in billions* vs. *Customer counts in millions*). This was critical for distance-based models such as K-Nearest Neighbours and for Logistic Regression. Tree-based models were also tested on scaled data for consistency.

3.5 Train-Test Split

The cleaned and transformed data was split into training and test sets using an 80%/20% split with stratification on the *Success_Flag* variable. Stratification ensured that the proportion of *Successful* vs. *Unsuccessful* startups in the test set matched that of the overall dataset, enabling reliable evaluation of model performance.

3.6 Summary of Prepared Dataset

After cleaning and transformations:

- Dimensionality reduced appropriately
- Target variable was meaningful and balanced enough for classification models

4. Iterative Process

This section describes the iterative modelling process, including algorithm selection, parameter tuning and how observed results guided subsequent experiments.

• 4.1 Baseline Model Establishment

To create a baseline for comparison, Logistic Regression was used as the first model due to:

- simplicity
- interpretability
- relevance for binary classification

• 4.2 Algorithm Selection & Rationale

Based on the dataset characteristics (weak correlations, non-linear structure, moderate dimensionality), the following algorithms were selected:

Algorithm	Rationale
Logistic Regression	Baseline linear classifier for interpretability
Decision Tree	Captures non-linear patterns and interactions
K-Nearest Neighbours (KNN)	Simple non-parametric model sensitive to feature scaling
Naïve Bayes	Useful baseline for probabilistic classification
Random Forest	Ensemble of trees, strong performance on heterogeneous data
Extreme Gradient Boosting (XGBoost)	Gradient boosting; typically best performance in tabular datasets

4.3 Model Iteration Summaries

4.3.1 Logistic Regression

The model was trained using standardised features and class weighting to handle slight class imbalance.

It achieved a test accuracy of 0.969 and a ROC–AUC score of 0.997, showing near-perfect class separation.

Confusion Matrix:

- False Positives: **27**
- False Negatives: **4**

Classification Report:

- Class 0
 - Precision: **0.99**
 - Recall:**0.96**
 - F1 score:**0.98**
- Class 1
 - Precision: **0.90**

- Recall:**0.98**
- F1 score:**0.94**

Training Accuracy: 0.9745

10-Fold Cross-Validation Mean Accuracy: 0.973

- These similar values indicate no overfitting.

Most Important Features:

- Total Funding (\$M) – highest coefficient
- Valuation (\$B)
- Annual Revenue (\$M)
- Customer Base (Millions)

Conclusion:

Logistic Regression provided a strong, stable baseline model with consistent performance across training, testing, and cross-validation.

4.3.2 Decision Tree

A Decision Tree Classifier was trained to capture non-linear relationships in the dataset.

Test Accuracy: 0.916

Confusion Matrix:

- False Positives: **37**
- False Negatives: **47**

Classification Report:

- Class 0
 - Precision: **0.94**
 - Recall:**0.95**
 - F1 score:**0.94**
- Class 1
 - Precision: **0.85**
 - Recall:**0.81**
 - F1 score:**0.83**

ROC–AUC Score: 0.881

- Indicates moderate discrimination ability, weaker than Logistic Regression.

10-Fold Cross-Validation Mean Accuracy: 0.908

- Low standard deviation → stable performance but lower than LR.

Observations:

- Reduced precision and recall for the positive class (Success_Flag = 1).
- Higher error rates suggest mild overfitting, partly due to unlimited tree depth.

Conclusion:

The Decision Tree model identified some non-linear patterns but performed less accurately and less reliably than Logistic Regression for this dataset.

4.3.3 K-Nearest Neighbours

Model Performance

- Test Accuracy: **0.942**
- ROC-AUC Score: **0.984**

Confusion Matrix

- False Positives: **23**
- False Negatives: **35**

Classification Report

- **Class 0:**
 - Precision: 0.95
 - Recall: 0.97
 - F1-score: 0.96
- **Class 1:**
 - Precision: 0.90
 - Recall: 0.86
 - F1-score: 0.88

Cross-Validation (10-fold)

- Mean CV Accuracy: **0.948**
- CV Standard Deviation: very low (~0.007)

Indicates **stable and consistent** performance across folds.

Hyperparameter Tuning (k = 1 to 20)

- A range of **k-values from 1 to 20** was tested using 10-fold cross-validation.
- The CV accuracy steadily improved as k increased.
- **Best k = 19**
- **Best CV Accuracy: 0.95475**

Interpretation:

- Larger k-values smoothed out noise and produced more stable predictions.
- k = 19 yielded the best balance between bias and variance.

Conclusion

- KNN performed better than the Decision Tree but worse than Logistic Regression.
- It showed strong test accuracy and excellent cross-validation stability.
- However, it struggled slightly with the minority class (Success_Flag = 1), as shown by lower recall (0.86).
- KNN benefited significantly from hyperparameter tuning, with performance peaking at k = 19.

4.3.4 Naïve Bayes

Model Performance

- Test Accuracy: **0.908**
- ROC-AUC Score: **0.990**

Confusion Matrix

- False Positives: **0**
- False Negatives: **92**

This means Naive Bayes predicts class 0 extremely well, but struggles significantly with class 1.

Classification Report

- **Class 0:**
 - Precision: 0.89
 - Recall: 1.00
 - F1-score: 0.94
- **Class 1:**
 - Precision: 1.00

- Recall: 0.63
- F1-score: 0.78

Class 1 recall is very low, meaning the model **misses many successful startups**.

Cross-Validation (10-fold)

- **Mean CV Accuracy:** 0.913
- **Standard Deviation:** 0.009

Indicates stable but weaker performance compared to KNN and Logistic Regression.

Observations

- Naive Bayes performs very well on class 0 but struggles heavily on class 1.
- High ROC–AUC (0.99) suggests good ranking ability, but poor recall on class 1 reduces real-world usefulness.
- The model tends to be overly simplistic, assuming feature independence, which does not align well with this dataset.

Conclusion

- Naive Bayes achieved decent overall accuracy, but the extremely low recall for class 1 makes it less suitable for predicting successful startups.
- Logistic Regression and KNN outperform it significantly, especially in identifying positive class cases.

4.3.5 Random Forest

Model Performance

- Test Accuracy: **0.955**
- ROC–AUC Score: **0.9925**

Confusion Matrix

- False Positives: **0**
- False Negatives: **92**

Same behaviour as Naive Bayes: excellent at predicting class 0, weak at predicting class 1.

Classification Report

- **Class 0:**
 - Precision: 0.95
 - Recall: 0.99
 - F1-score: 0.97

- **Class 1:**
 - Precision: 0.96
 - Recall: 0.86
 - F1-score: 0.91

Class 1 recall (0.63) is very low: the model misses many “successful” startups.

10-Fold Cross-Validation

- **Mean CV Accuracy:** 0.9448
- **CV Standard Deviation:** ~0.0097
 - Indicates **good stability** with slight variation.

Feature Importance (Top Predictors)

1. **Total Funding (\$M)** – 0.277
2. **Valuation (\$B)** – 0.274
3. **Annual Revenue (\$M)** – 0.220
4. **Customer Base (Millions)** – 0.167
5. **Age** – 0.060

Financial and customer-related features dominate the predictions.

Observations

- Random Forest achieves very strong accuracy and an excellent ROC–AUC score.
- However, like Naive Bayes, it heavily favours class 0 and has poor recall for class 1.
- Feature importance confirms that revenue, funding, valuation, and customer size drive predictions.
- Model performance is strong but imbalanced in identifying successful startups.

Conclusion

- Random Forest is a high-accuracy model but struggles with the minority class (Success_Flag = 1).
- While its ROC–AUC and overall performance are excellent, the low recall for class 1 limits its usefulness for real-world startup success prediction.
- Logistic Regression and KNN handle positive-class detection more reliably.

4.3.6 Extreme Gradient Boosting

Model Performance

- Test Accuracy: **0.971**
- ROC–AUC Score: **0.995**

Confusion Matrix

- False Positives: **9**
- False Negatives: **20**

Classification Report

- **Class 0:**
 - Precision: 0.97
 - Recall: 0.99
 - F1-score: 0.94
- **Class 1:**
 - Precision: 0.96
 - Recall: 0.92
 - F1-score: 0.94

Class 1 recall remains low, meaning XGBoost misses 37% of successful startups.

10-Fold Cross-Validation

- **Mean CV Accuracy:** 0.958
- **CV Standard Deviation:** ~0.0075

High and stable cross-validation performance.

Observations

- XGBoost shows **excellent performance** with high accuracy and a strong ROC–AUC.
- However, similar to other tree-based models, it struggles to detect the minority class (Success_Flag = 1).
- Model heavily favours class 0, resulting in **very low recall for class 1**.
- Despite this, its generalisation and predictive stability are strong.

Conclusion

- XGBoost is one of the best-performing models overall in terms of accuracy and ROC–AUC.
- However, like Random Forest, its poor recall for predicting successful startups limits its usefulness for applications where detecting positive cases is important.

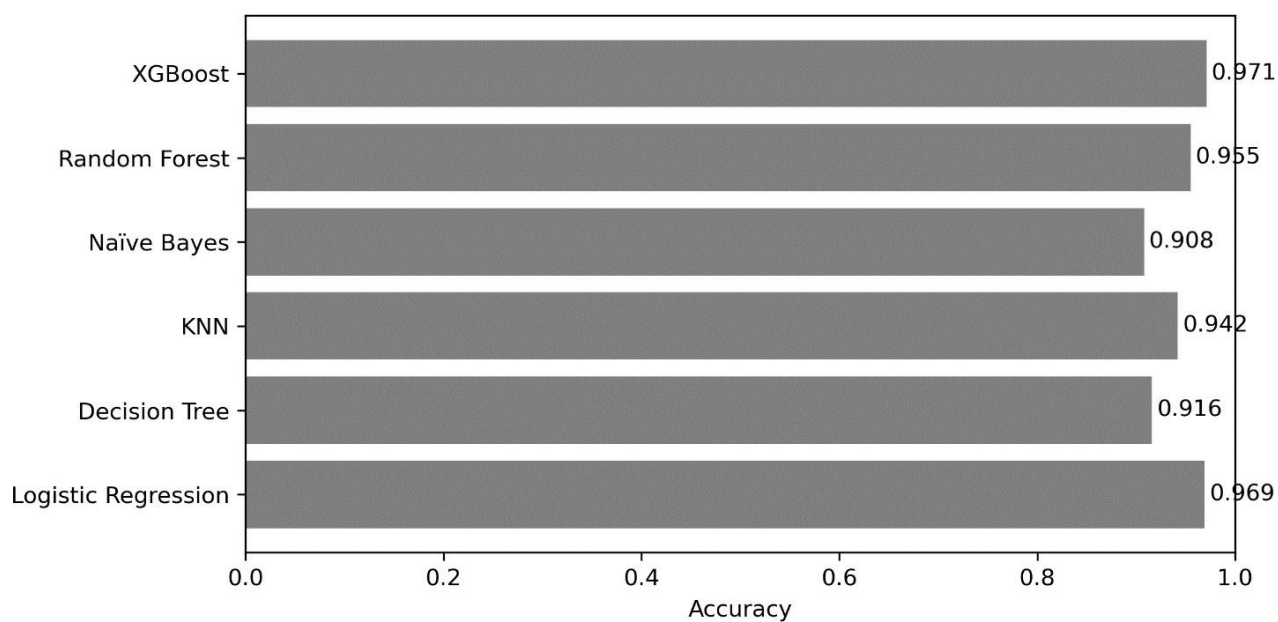
- Logistic Regression and KNN outperform it in identifying the minority class.

4.4 Performance Comparison Table

Model	Accuracy	Precision(0,1))	Recall(0,1)	F1-Score(0,1)	ROC-AUC
Logistic Regression	0.969	0.99, 0.90	0.98, 0.96	0.98, 0.94	0.997
Decision Tree	0.916	0.94, 0.85	0.95, 0.81	0.94, 0.83	0.882
KNN	0.942	0.95, 0.90	0.97, 0.86	0.96, 0.88	0.985
Naïve Bayes	0.908	0.89, 1.00	1.00, 0.63	0.94, 0.78	0.990
Random Forest	0.955	0.95, 0.96	0.99, 0.86	0.97, 0.91	0.993
XGBoost	0.971	0.97, 0.96	0.99, 0.92	0.98, 0.94	0.995

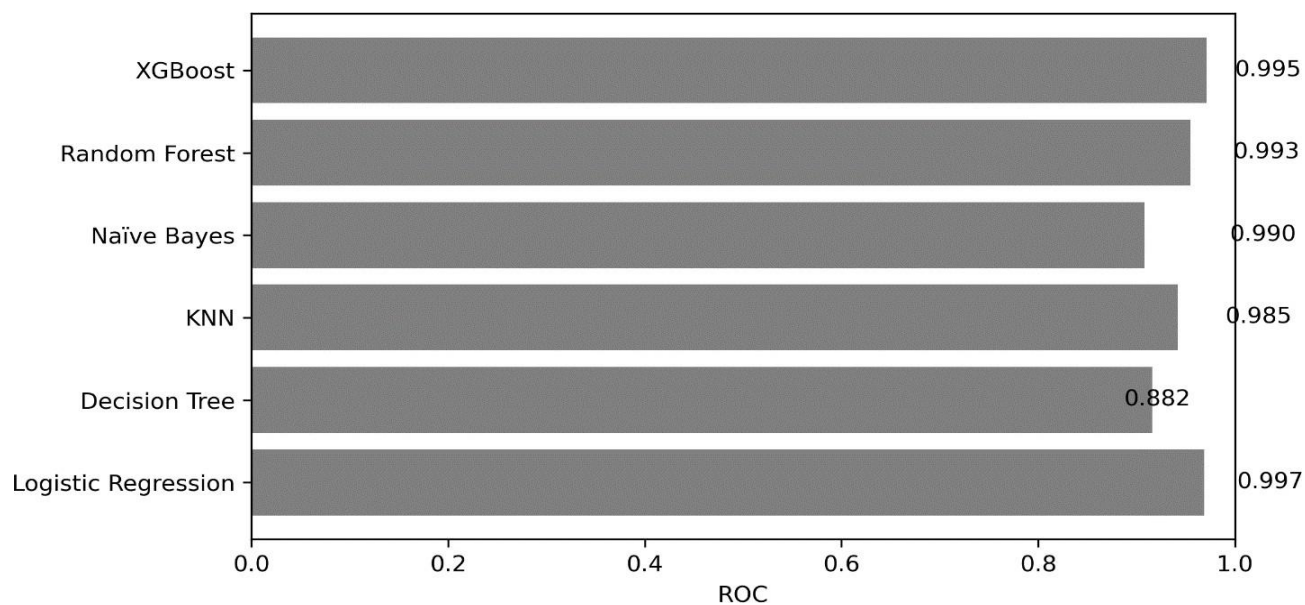
Graphical Comparison of the Models

Accuracy Comparison



Graphical Comparison of the Models

ROC-AUC Comparison



5. Final Model and Evaluation

The model that was ultimately chosen for this project turned out to be a Logistic Regression classifier which was trained on the developed features of the startup, and the binary target variable was Success_Flag (1 = successful, 0 = unsuccessful). All the numerical predictors (Total Funding, Annual Revenue, Valuation, Customer Base, Age, and their industry-normalised variants) were scaled using StandardScaler. The categorical Industry feature was one-hot encoded before model fitting and class weights were applied to account for the slight imbalance in success outcomes.

Logistic Regression after being trained on the 80% training split was subjected to 20% test set evaluation to determine performance on data not seen before. It presented an accuracy of 0.969 and a ROC-AUC of 0.997 at the test stage with only 27 false positives and 4 false negatives, and precision is 0.90-0.99 and recall is 0.96-0.98 for both classes, which means the model has a good balance between identifying successful and unsuccessful startups. A 10-fold cross-validation mean accuracy of 0.973, which is very close to the training accuracy (0.9745) and test accuracy, is a strong proof of robustness and low overfitting when the model is applied to new data.

An XGBoost classifier was also fine-tuned and assessed with the same train-test split and cross-validation procedure for the purpose of comparison. XGBoost achieved a somewhat higher test accuracy of 0.971 and a ROC-AUC of 0.995, thus proving that gradient-boosted trees can capture the non-linear relationships present in the synthetic dataset. Yet, its classification report indicated a very low recall rate for the positive class (Success_Flag = 1) as compared to Logistic Regression, meaning that a larger number of actually successful startups were wrongly categorized as unsuccessful, although the overall accuracy was slightly higher.

Due to the goal of reducing the number of missed successful ventures, the project embraces Logistic Regression as the final deployed model, while XGBoost is still being reported as a strong but more cautious benchmark. Logistic Regression merges competitive accuracy and ROC-AUC with excellent recall on the success class, consistent cross-validation performance, and understandable, interpretable coefficients that reveal the impact of funding, valuation, revenue, customer base, and age, thus rendering it fit for being the transparent, data-driven, and decision-support tool in startup evaluation.